

ACR15044

## 4-Chlorobenzaldehyde

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 4-氯苯甲醛  
4-Chlorobenzaldehyde

**Cat No. :** 150440000; 150440010; 150441000; 150442500  
**Synonyms** p-Chlorobenzencarboxaldehyde; p-Chlorobenzaldehyde  
**CAS No** 104-88-1  
**Molecular Formula** C7 H5 Cl O

**Supplier**

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticlaan 3a, 2440 Geel, Belgium

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Solid

**Appearance**  
White to yellow

**Odor**  
No information available

#### Emergency Overview

Harmful if swallowed. Causes skin irritation. May cause an allergic skin reaction. Causes serious eye irritation. Toxic to aquatic life with long lasting effects. Sensitivity to light. Air sensitive.

#### Classification of the substance or mixture

|                                   |            |
|-----------------------------------|------------|
| Acute Oral Toxicity               | Category 4 |
| Skin Corrosion/Irritation         | Category 2 |
| Serious Eye Damage/Eye Irritation | Category 2 |
| Skin Sensitization                | Category 1 |
| Chronic aquatic toxicity          | Category 2 |

#### Label Elements

## 4-Chlorobenzaldehyde



## Signal Word

## Warning

## Hazard Statements

H302 - Harmful if swallowed  
H315 - Causes skin irritation  
H317 - May cause an allergic skin reaction  
H319 - Causes serious eye irritation  
H411 - Toxic to aquatic life with long lasting effects

## Precautionary Statements

## Prevention

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Response

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P330 - Rinse mouth  
P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention  
P337 + P313 - If eye irritation persists: Get medical advice/attention  
P362 + P364 - Take off contaminated clothing and wash it before reuse

## Storage

P403 - Store in a well-ventilated place

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

None identified.

## Health Hazards

Harmful if swallowed. Causes skin irritation. May cause an allergic skin reaction. Causes serious eye irritation.

## Environmental hazards

Toxic to aquatic life with long lasting effects. . Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

## Other Hazards

May form explosible dust-air mixture if dispersed. Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component            | CAS No   | Weight % |
|----------------------|----------|----------|
| p-Chlorobenzaldehyde | 104-88-1 | >95      |

## SECTION 4. FIRST AID MEASURES

## General Advice

If symptoms persist, call a physician.

## 4-Chlorobenzaldehyde

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Combustible material. Containers may explode when heated.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Avoid dust formation. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Remove all sources of ignition.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

## 4-Chlorobenzaldehyde

Wear personal protective equipment/face protection. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Ensure adequate ventilation. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Keep under nitrogen.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust MDHS70 General methods for sampling airborne gases and vapours

**Exposure Controls****Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection**

Goggles (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use**

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use**

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Particle filtering: EN149:2001

When RPE is used a face piece Fit Test should be conducted

## 4-Chlorobenzaldehyde

|                                 |                                                                                                 |
|---------------------------------|-------------------------------------------------------------------------------------------------|
| Hygiene Measures                | Handle in accordance with good industrial hygiene and safety practice.                          |
| Environmental exposure controls | Prevent product from entering drains. Do not allow material to contaminate ground water system. |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                         |                          |                                        |
|-----------------------------------------|--------------------------|----------------------------------------|
| Appearance                              | White to yellow          |                                        |
| Physical State                          | Solid                    |                                        |
| Odor                                    | No information available |                                        |
| Odor Threshold                          | No data available        |                                        |
| pH                                      | No information available |                                        |
| Melting Point/Range                     | > 46 °C / > 114.8 °F     |                                        |
| Softening Point                         | No data available        |                                        |
| Boiling Point/Range                     | 214 °C / 417.2 °F        |                                        |
| Flash Point                             | 88 °C / 190.4 °F         | Method - No information available      |
| Evaporation Rate                        | Not applicable           | Solid                                  |
| Flammability (solid,gas)                | No information available |                                        |
| Explosion Limits                        | No data available        |                                        |
| Vapor Pressure                          | 0.36 mbar @ 25 °C        |                                        |
| Vapor Density                           | Not applicable           | Solid                                  |
| Specific Gravity / Density              | 1.196                    |                                        |
| Bulk Density                            | No data available        |                                        |
| Water Solubility                        | 935 mg/L (20°C)          |                                        |
| Solubility in other solvents            | No information available |                                        |
| Partition Coefficient (n-octanol/water) |                          |                                        |
| Component                               | log Pow                  |                                        |
| p-Chlorobenzaldehyde                    | 2.19                     |                                        |
| Autoignition Temperature                | 250 °C / 482 °F          |                                        |
| Decomposition Temperature               | No data available        |                                        |
| Viscosity                               | Not applicable           | Solid                                  |
| Explosive Properties                    |                          | explosive air/vapour mixtures possible |
| Oxidizing Properties                    | No information available |                                        |
| Molecular Formula                       | C7 H5 Cl O               |                                        |
| Molecular Weight                        | 140.57                   |                                        |

## SECTION 10. STABILITY AND REACTIVITY

|                          |                                                                                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stability                | Air sensitive. Light sensitive.                                                                                                                                             |
| Hazardous Reactions      | None under normal processing.                                                                                                                                               |
| Hazardous Polymerization | No information available.                                                                                                                                                   |
| Conditions to Avoid      | Avoid dust formation. Incompatible products. Heat, flames and sparks. Exposure to air. Exposure to light. Keep away from open flames, hot surfaces and sources of ignition. |
| Materials to avoid       | Strong oxidizing agents. Strong reducing agents. Strong bases.                                                                                                              |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## 4-Chlorobenzaldehyde

## (a) acute toxicity;

| Component            | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation               |
|----------------------|---------------------------|---------------------------|-------------------------------|
| p-Chlorobenzaldehyde | LD50 = 1575 mg/kg ( Rat ) | LD50 > 5000 mg/kg ( Rat ) | LC50 > 0.321 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 2

## (d) respiratory or skin sensitization;

Respiratory No data available  
Skin Category 1  
No information available

(e) germ cell mutagenicity; No data available  
Not mutagenic in AMES Test

(f) carcinogenicity; No data available  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; Not applicable  
Solid

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## SECTION 12. ECOLOGICAL INFORMATION

## Ecotoxicity effects

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component            | Freshwater Fish                                            | Water Flea | Freshwater Algae | Microtox                                                                                                                        |
|----------------------|------------------------------------------------------------|------------|------------------|---------------------------------------------------------------------------------------------------------------------------------|
| p-Chlorobenzaldehyde | LC50: 2 - 2.4 mg/L, 96h flow-through (Pimephales promelas) |            |                  | EC50 = 10.1 mg/L 30 min<br>EC50 = 10.2 mg/L 5 min<br>EC50 = 10.4 mg/L 30 min<br>EC50 = 10.9 mg/L 15 min<br>EC50 = 500 mg/L 16 h |

Persistence and Degradability Not readily biodegradable

**SAFETY DATA SHEET****4-Chlorobenzaldehyde****Persistence  
Degradation in sewage  
treatment plant**

Persistence is unlikely.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

Bioaccumulation is unlikely

| Component            | log Pow | Bioconcentration factor (BCF) |
|----------------------|---------|-------------------------------|
| p-Chlorobenzaldehyde | 2.19    | No data available             |

**Mobility in soil**

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information  
Persistent Organic Pollutant  
Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused  
Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

UN-No UN3077  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s.  
Technical Shipping Name (4-CHLOROBENZALDEHYDE)  
Hazard Class 9  
Packing Group III

**IMDG/IMO**

UN-No UN3077  
Proper Shipping Name Environmentally hazardous substances, solid, n.o.s.  
Technical Shipping Name (4-CHLOROBENZALDEHYDE)  
Hazard Class 9  
Packing Group III

**IATA**

UN-No UN3077  
Proper Shipping Name ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.\*  
Technical Shipping Name (4-CHLOROBENZALDEHYDE)  
Hazard Class 9  
Packing Group III

**Special Precautions for User**

No special precautions required

**SECTION 15. REGULATORY INFORMATION**

## 4-Chlorobenzaldehyde

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component            | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|----------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| p-Chlorobenzaldehyde | -                                                   | -                                       | X    | X     | 203-247-4 | X    | -   | X     | X    | X    | X    | KE-05521 |

## National Regulations

## SECTION 16. OTHER INFORMATION

**Creation Date** 18-Oct-2010  
**Revision Date** 06-Apr-2024  
**Revision Summary** SDS sections updated.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

First aid for chemical exposure, including the use of eye wash and safety showers.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS



---

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**